



NORTH SYDNEY BOYS HIGH SCHOOL

2011 HSC ASSESSMENT TASK 3

Mathematics Extension 2

Examiner: S. Ireland

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write on both sides of the paper (with lines) in the booklet(s) provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- This is a school assessment task. The task's content, format and mark scheme do not necessarily reflect that of the HSC.

- Attempt all questions
- Each new question is to be started on a **new page**.

Class Teacher:

(Please tick or highlight)

- ☐ Ms Collins
- ☐ Mr Fletcher
- ☐ Mr Ireland

Student Number

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	7	8	Total	Total
Mark	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{120}$	$\overline{100}$

Question 1 (15 marks) Start a new page. **Marks**

(a) Evaluate $\int_1^3 \frac{dx}{x(x+2)}$ 3

(b) Evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x \tan^4 x \, dx$ 3

(c) Find $\int \frac{x}{x^2 + 2x + 5} dx$ 3

(d) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x + \cos x}$ 4

(e) Find $\int \frac{dx}{\sqrt{x^2 - 6x + 8}}$ 2

Question 2 (15 marks) Start a new page.

(a) If $z = \frac{1+7i}{3-4i}$, then

(i) Write z in $a + ib$ form (where a and b are real) 1

(ii) Find $|z|$ 1

(iii) Find $\arg z$ 1

(iv) Calculate z^8 1

(b) On separate Argand diagrams sketch the locus of a point which satisfies:

(i) $\arg(z + 1 + i) = \frac{\pi}{4}$ 1

(ii) $\operatorname{Re}(z) + \operatorname{Im}(\bar{z}) = 1$ 1

(c) Express the square root of $-2i$ in the form $a + ib$ 2

(d) If z is a complex number such that $z + \frac{1}{z}$ is real, prove that either z is real or $|z| = 1$. 3

(e) The equation $z^3 + az^2 + bz + 6 = 0$, where a and b are real numbers, has $1 + i$ as a root. 4

Find a and b , and solve the equation completely.

Question 3 (15 marks) Start a new page.**Marks**

- (a) (i) Prove that for any polynomial $P(x)$, if k is a zero of multiplicity r , then k is a zero of multiplicity $r - 1$ of $P'(x)$. 1

- (ii) Given that the polynomial $P(x) = x^4 + 5x^3 + 9x^2 + 7x + 2$ has a zero of multiplicity 3, factorise $P(x)$. 3

- (b) The equation $x^3 - 4x^2 + 5x + 2 = 0$ has roots α, β, γ .

Find:

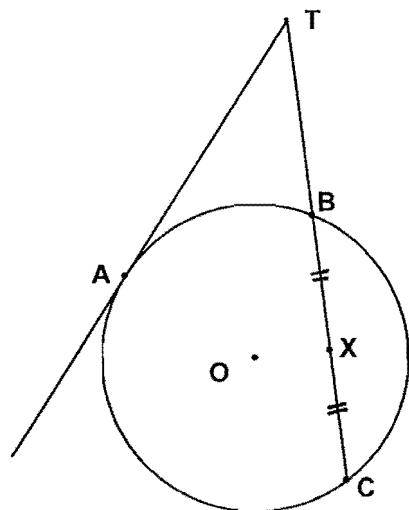
(i) $\alpha^2 + \beta^2 + \gamma^2$ 2

(ii) $\alpha^3 + \beta^3 + \gamma^3$ 2

- (c) If α, β , and γ are roots of $8x^3 - 4x^2 + 6x - 1 = 0$, find the equation whose roots are $\frac{1}{1-\alpha}, \frac{1}{1-\beta}$ and $\frac{1}{1-\gamma}$. 3

- (d)

- (i) Copy the diagram into your answer booklet:

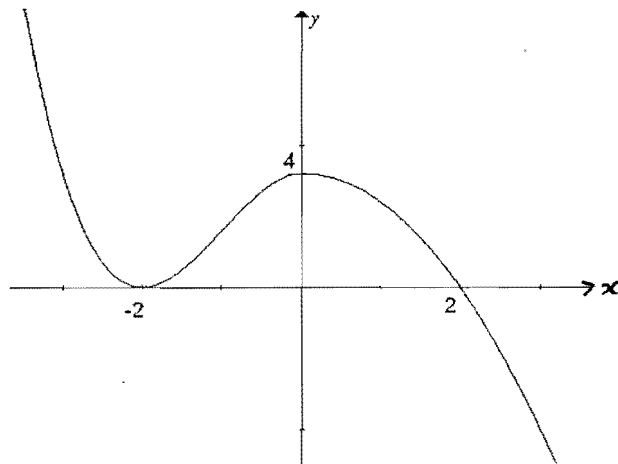


- (ii) A, B, C are three points on the circumference of a circle, centre O . The tangent at A meets BC produced at T . X is the midpoint of BC . Prove that $\angle AOT = \angle AXT$. 4

Question 4 (15 marks) Start a new page.

Marks

(a) The diagram shows $y = f(x)$.



Draw separate one-third page sketches of the following:

- | | | |
|-------|----------------------|---|
| (i) | $y = \frac{1}{f(x)}$ | 2 |
| (ii) | $y = f(x) $ | 1 |
| (iii) | $ y = f(x)$ | 2 |
| (iv) | $y^2 = f(x)$ | 2 |
| (v) | $y = e^{f(x)}$ | 2 |

(b) Find the equation of the tangent to $x^3 + xy - y^3 = 1$ at the point $(1, 1)$. 3

(c) Sketch on the same number plane $y = |x| - 2$ and $y = 4 + 3x - x^2$.

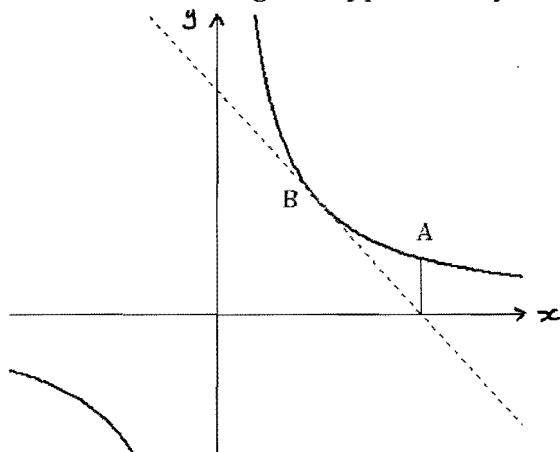
Hence, or otherwise, solve $\frac{|x|-2}{4+3x-x^2} > 0$ 3

Question 5 (15 marks) Start a new page.	Marks
(a) Let $P(x_1, y_1)$ be a point on the ellipse $16x^2 + 25y^2 = 400$.	
(i) Draw the ellipse, showing all intercepts.	1
(ii) Write down the eccentricity.	1
(iii) Show that the normal at P has equation	
$25y_1x - 16x_1y - 9x_1y_1 = 0$	2
(iv) The normal at P meets the major axis at N . Using the focus-directrix definition of an ellipse, or otherwise,	
prove that $\frac{NS}{NS'} = \frac{PS}{PS'}$	3
(b) The area enclosed by the parabola $y = (x - 3)^2$ and the straight line $y = 9$ is rotated about the y -axis. Use the method of cylindrical shells to find the exact volume of the resulting solid.	4
(c) The region bounded by the curve $y = x^2$ and the straight line $y = 4$ is rotated about the line $x = 2$. Use a slicing method – with slices perpendicular to the axis of rotation – to find the exact volume of the resulting solid.	4

Question 6 (15 marks) Start a new page.

Marks

- (a) A and B are variable points on the rectangular hyperbola $xy = c^2$.



The tangent at B passes through the foot of the ordinate (y -value) of A .

- | | | |
|------|---|---|
| (i) | If A and B have parameters t_1 and t_2 show that $t_1 = 2t_2$. | 3 |
| (ii) | Hence prove that the locus of the midpoint of AB is also a rectangular hyperbola. | 2 |

- (b) (i) Show that the locus specified by

$$|z - 2| = 2 \left(\operatorname{Re}(z) - \frac{1}{2} \right)$$

is a branch of the hyperbola $\frac{x^2}{1} - \frac{y^2}{3} = 1$, and indicate why it must be

that particular branch. 3

- | | | |
|------|---|---|
| (ii) | Sketch the locus, and find the possible set of values of each of $ z $ and $\arg z$ for a point on the locus. | 3 |
|------|---|---|

- (c) The region bounded by the parabolas $y = 6 - x^2$ and $y = \frac{1}{2}x^2$ forms the base of a solid. Cross-sections by planes perpendicular to the y -axis are semi-circles, with their diameters in the base of the solid.

- | | | |
|------|---|---|
| (i) | Find the points of intersection of the two parabolas. | 1 |
| (ii) | Find the volume of the solid. | 3 |

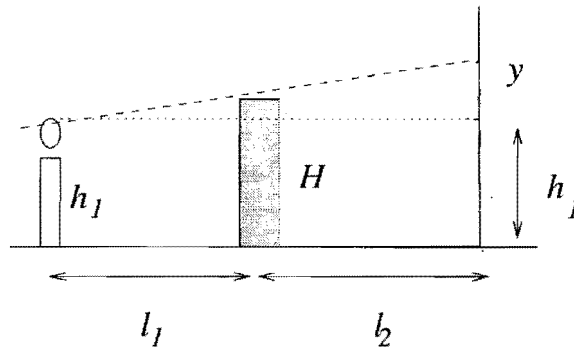
Question 7 (15 marks) Start a new page.

Marks

- (a) A castle with walls 30 metres high is surrounded by a moat 20 metres wide. An archer, kneeling at the edge of the moat, attempts to shoot over the wall.

- (i) Taking g as 10 ms^{-2} and the speed of an arrow as it leaves the bowstring as 40 ms^{-1} - and disregarding air resistance - derive the equations of motion for an arrow. 2
- (ii) Calculate the range of angles through which the archer must fire in order to clear the top of the wall. 4

- (b) A candle is placed a distance l_1 from a thin block of wood of height H . The block is a distance l_2 from a wall, as shown in the diagram:



The candle burns down so that the height of the flame h_1 decreases at the rate of 3 cm/h . Find the rate at which the length of the shadow, y , cast by the block on the wall increases.

(Your answer will be in terms of the constants l_1 and l_2). 3

- (c) For what value of k does the equation $e^{2x} = k\sqrt{x}$ have exactly one solution? 3

- (d) The function f is defined by

$$f(x) = \sqrt{8x - x^2} - \sqrt{14x - x^2 - 48}.$$

Find the maximum value of $f(x)$ using a graphical method. 3

Question 8 (15 marks) Start a new page.

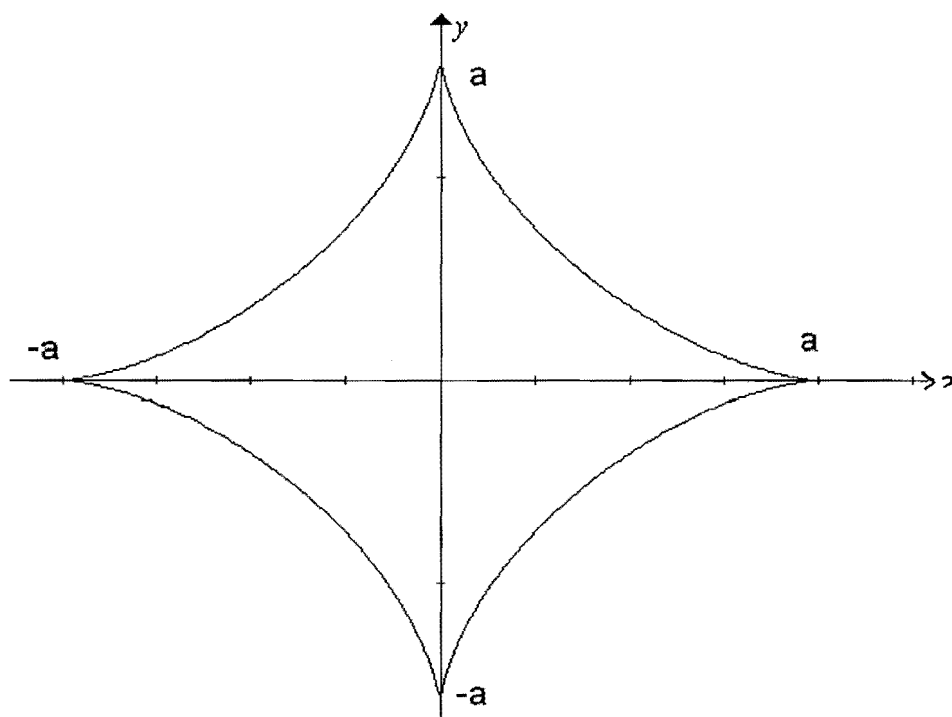
Marks

- (a) If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$, show that $I_n = \frac{n-1}{n} I_{n-2}$.

Hence evaluate I_4 and I_6 .

5

- (b) The asteroid curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ is sketched below:



- (i) Show that a parametric representation of the curve is given by
 $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ 1
- (ii) Show that the gradient of the tangent to the asteroid at any point
 $P(a \cos^3 \phi, a \sin^3 \phi)$ on it is equal to $\frac{dy}{dx} = -\tan \phi$. 1
- (iii) Show that the length of a tangent line to the asteroid at any point
 $P(a \cos^3 \phi, a \sin^3 \phi)$ on it, cut off by the coordinate axes, is constant. 3
- (iv) Find the area enclosed by the asteroid curve. [The substitution
 $x = a \cos^3 \theta$ and the results of Question 8 part (a) may be useful.] 5

END OF EXAMINATION

1 (a)

$$\text{Let } \frac{1}{x(x+2)} = \frac{a}{x} + \frac{b}{x+2}$$

$$\therefore 1 = (a+b)x + 2a$$

$$\therefore \begin{cases} a = \frac{1}{2} \\ b = -\frac{1}{2} \end{cases}$$

$$\therefore \int_1^3 \frac{dx}{x(x+2)} = \frac{1}{2} \int_1^3 \left(\frac{1}{x} - \frac{1}{x+2} \right) dx$$

$$= \frac{1}{2} \left[\ln x - \ln(x+2) \right]_1^3$$

$$= \frac{1}{2} \left[(\ln 3 - \ln 5) - (\ln 1 - \ln 3) \right]$$

$$= \frac{1}{2} [2 \ln 3 - \ln 5]$$

$$= \frac{1}{2} \ln \left(\frac{9}{5} \right)$$

Correct Solution : 3 with working

Applies partial fractions correctly but does not get answer correctly. : 2

Reasonable attempt to use partial fractions : 1

Q1 ctd

$$(c) \int \frac{x}{x^2+2x+5} dx = \int \frac{\frac{1}{2}(2x+2)-1}{x^2+2x+5} dx$$

ALT: Complete square in denominator & use $u = x+1$ substitution.

$$= \frac{1}{2} \int \frac{2x+2}{x^2+2x+5} dx - \int \frac{dx}{(x+1)^2+4}$$

$$= \frac{1}{2} \ln(x^2+2x+5) - \frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) + C$$

✓ formats numerator appropriately

✓ correctly split into 2 integrals

✓ correct answer

$$(d) \text{ Evaluate } \int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x + \cos x}$$

$$\text{Let } t = \tan \frac{x}{2} \therefore \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (1+t^2)$$

$$\therefore dx = \frac{2 dt}{1+t^2}$$

$$\text{Also, when } x=0, t=0 \text{ \& } x=\frac{\pi}{2}, t=1$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x + \cos x} = \int_0^1 \frac{2 dt}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$$

$$= \int_0^1 \frac{2 dt}{1+t^2+2t+1-t^2}$$

$$= \int_0^1 \frac{1}{1+t} dt$$

$$= [\ln(1+t)]_0^1$$

$$= \ln 2$$

✓ Correctly sets up & substitution

✓ Correct substitution in integral

✓✓ Correct working and answer

1 (b)

$$\int_0^{\frac{\pi}{4}} \sec^4 x \tan^4 x dx = \int_0^{\frac{\pi}{4}} (1+\tan^2 x) \cdot \tan^4 x \cdot \sec^2 x dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan^4 x + \tan^6 x) \sec^2 x dx$$

$$= \left[\frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} \right]_0^{\frac{\pi}{4}}$$

$$= \left(\frac{1}{5} + \frac{1}{7} \right) - (0+0)$$

$$= \frac{12}{35}$$

✓ recognises & uses trig. identity correctly

✓ correct primitive

✓ correct answer

ctd. 1 (c)

$$\int \frac{dx}{\sqrt{x^2-6x+8}} = \int \frac{dx}{\sqrt{(x-3)^2-1}}$$

$$= \ln \left| x-3 + \sqrt{x^2-6x+8} \right| + C$$

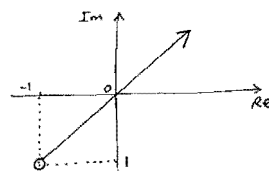
✓ complete square

✓ correct answer

[no penalty for omitting "C" or absolute value signs.]

Q2 ctd

$$(b) (i) \arg(z - (-1-i)) = \frac{\pi}{4}$$

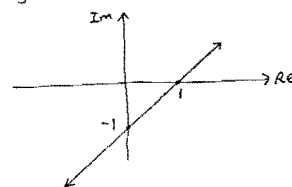


✓ correct sketch

$$(ii) \operatorname{Re}(z) + \operatorname{Im}(z) = 1$$

$$\text{Let } z = x+iy \therefore \bar{z} = x-iy$$

$$\therefore x-y=1$$



✓ correct sketch

2

$$(i) \frac{1+7i}{3-4i} \times \frac{3+4i}{3+4i} = \frac{3+4i+21i-28}{3^2+4^2}$$

$$= \frac{-25+25i}{25}$$

$$= -1+i$$

✓ correct answer

$$(ii) |z| = \sqrt{1^2+1^2} = \sqrt{2}$$

✓ correct answer

$$(iii) \arg z = \frac{3\pi}{4}$$

✓ correct answer

$$(iv) z^8 = \left[\sqrt{2} \operatorname{cis} \frac{3\pi}{4} \right]^8 = (\sqrt{2})^8 \operatorname{cis} \left(8 \cdot \frac{3\pi}{4} \right) = 16 \operatorname{cis} 6\pi = 16$$

✓ correct answer

$$(c) (a+ib)^2 = -2i$$

$$(a^2-b^2) + (2ab)i = 0 - 2i$$

$$\therefore \begin{cases} a^2-b^2=0 \\ ab=-1 \end{cases}$$

$$b = -\frac{1}{a} \therefore a^2 - \frac{1}{a^2} = 0$$

$$a^4 - 1 = 0$$

$$(a^2-1)(a^2+1) = 0$$

$$\therefore a = -1 \text{ or } 1 \quad (a \text{ is real})$$

$$\therefore b = 1 \text{ or } -1$$

Thus square roots are $1-i$ and $-1+i$ (i.e. $\pm(1-i)$)

✓

ctd 2(d)

Let $z = x + iy$
 $\therefore z + \frac{1}{z} = x + iy + \frac{1}{x + iy} \cdot \frac{x - iy}{x - iy}$
 $= x + iy + \frac{x - iy}{x^2 + y^2}$
 $= \left(x + \frac{x}{x^2 + y^2}\right) + i\left(y - \frac{y}{x^2 + y^2}\right)$

if $z + \frac{1}{z}$ is real, then $y = \frac{y}{x^2 + y^2}$

$y(x^2 + y^2) = y$
 $y(x^2 + y^2 - 1) = 0$

hence $y = 0$ or $x^2 + y^2 - 1 = 0$
 i.e. z is real or $|z| = 1$.

(e) $z^3 + az^2 + bz + 6 = 0$

Since $1+i$ is a root, so is $1-i$
 (as coefficients are real).

Let the third root be γ .

Then $(1+i)(1-i)(\gamma) = -6$ [product of roots]
 $\therefore 2\gamma = -6$
 $\gamma = -3$

Also, since $(1+i) + (1-i) + (-3) = -a$ [sum of roots]
 $\therefore -1 = -a$
 $\therefore a = 1$

Finally, we have:

$(1+i)(-3) + (1-i)(-3) + (1+i)(1-i) = b$
 $\therefore -3-3i-3+3i+2 = b$
 $\therefore b = -4$

✓ Simplifies $z + \frac{1}{z}$ correctly

✓ uses assumption correctly

✓ Correct conclusion

✓ Correct conjugate root

✓ Correct third root

✓ correct a

✓ correct b

Q3

(a) (i) Let k be a zero of multiplicity $r > 1$
 for $P(x)$.

Then $P(x) = (x-k)^r \cdot Q(x)$ [and k is not a zero of $Q(x)$]

$\therefore P'(x) = Q(x) \cdot r(x-k)^{r-1} + (x-k)^r \cdot Q'(x)$

$= (x-k)^{r-1} [rQ(x) + (x-k) \cdot Q'(x)]$

$= (x-k)^{r-1} \cdot Q_1(x)$

where $Q_1(k) \neq 0$, as $Q(k) \neq 0$.

$\therefore k$ is a zero of multiplicity $r-1$ of $P'(x)$.

✓ Essentially correct proof.

(ii) $P(x) = x^4 + 5x^3 + 9x^2 + 7x + 2$

$\therefore P'(x) = 4x^3 + 15x^2 + 18x + 7$

$\therefore P''(x) = 12x^2 + 30x + 18$

$= 6(2x^2 + 5x + 3)$

$= 6(2x+3)(x+1)$

Thus, by (i), either $-\frac{3}{2}$ or -1 is a zero of multiplicity 3 of $P'(x)$.

Since $P'(-1) = P(-1) = 0$, it must be -1 .

$\therefore$ (by inspection) $P(x) = (x+1)^3(x+2)$.

✓ identifies possible zeros of multiplicity 3.

✓ determines it is -1

✓ final factorisation

Q3

ctd.

(b) $x^3 - 4x^2 + 5x + 2 = 0$

1) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= 4^2 - 2(5)$
 $= 6$

2) $\alpha^3 - 4\alpha^2 + 5\alpha + 2 = 0$
 $\beta^3 - 4\beta^2 + 5\beta + 2 = 0$
 $\gamma^3 - 4\gamma^2 + 5\gamma + 2 = 0$

$(\alpha^3 + \beta^3 + \gamma^3) - 4(\alpha^2 + \beta^2 + \gamma^2) + 5(\alpha + \beta + \gamma) + 6 = 0$

$\therefore \alpha^3 + \beta^3 + \gamma^3 = 4(6) - 5(4) - 6$
 $= -2$

✓ correct start

✓ correct answer

✓ correct idea of subtracting & adding

✓ correct answer

3) (c) $8x^3 - 4x^2 + 6x - 1 = 0$ has roots α, β, γ .

Let $y = \frac{1}{1-x}$

$\therefore 1-x = \frac{1}{y}$

$\therefore x = 1 - \frac{1}{y}$

$\therefore x = \frac{y-1}{y}$

Thus required equation is:-

$8\left(\frac{y-1}{y}\right)^3 - 4\left(\frac{y-1}{y}\right)^2 + 6\left(\frac{y-1}{y}\right) - 1 = 0$

$\therefore 8(y-1)^3 - 4y(y-1)^2 + 6y^2(y-1) - y^3 = 0$

$\therefore 8y^3 - 24y^2 + 24y - 8 - 4y^3 + 8y^2 - 4y + 6y^3 - 6y^2 - y^3 = 0$

$\therefore 9y^3 - 22y^2 + 20y - 8 = 0$

(i.e. $9x^3 - 22x^2 + 20x - 8 = 0$,
 or any multiple of this)

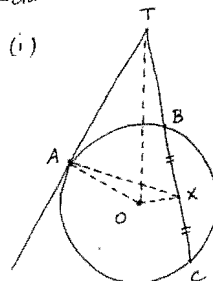
✓

✓

✓

Q3 ctd.

(d) (i)



(ii) Join OX, OA, OT, AX .

Then:

$\angle OXB = 90^\circ$ [Centre to midpoint of chord is perpendicular to it]

$\angle OAT = 90^\circ$ [tangent is $\perp$ to radius at point of contact]

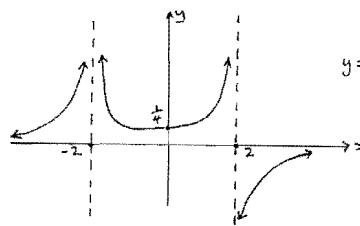
$\therefore OATX$ is a cyclic quadrilateral [opposite angles supplementary]

$\therefore \angle AOT = \angle AXT$ [angles standing on same arc are equal]

✓ [needs accurate reasons for full marks]

Q4

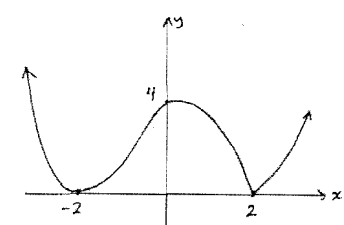
(a) (i)



$y = \frac{1}{f(x)}$

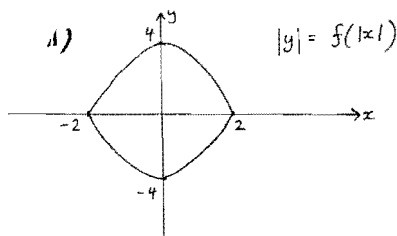
✓ ✓
 [asymptotes & intercept needed for full marks]

(ii)

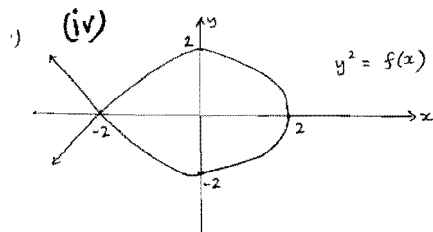


$y = |f(x)|$

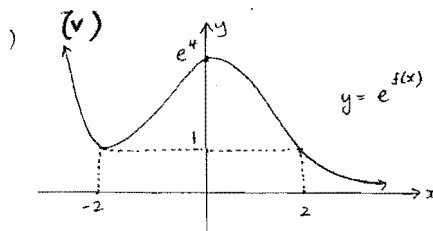
✓



✓✓
(note: is 'pointy' at $x = \pm 2$)



✓✓
(note: vertical tangent at $x = 2$)



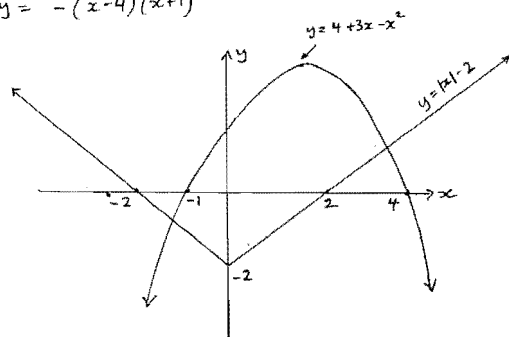
✓ basic shape + intercept
✓ correct behavior at $x = \pm 2$

(b) $x^3 + xy - y^3 = 1$
 $3x^2 + y + x \frac{dy}{dx} - 3y^2 \frac{dy}{dx} = 0$
 at (1,1) we have:-
 $3 + 1 + \frac{dy}{dx} - 3 \frac{dy}{dx} = 0$
 $\therefore \frac{dy}{dx} = 2$
 $\therefore y - 1 = 2(x - 1)$
 $\therefore y = 2x - 1$
 $\therefore 2x - y - 1 = 0$

✓ correct implicit differentiation
✓ correct value of $\frac{dy}{dx}$ at (1,1)
✓ correct final answer

Q4 - ctd

(c) $y = 4 + 3x - x^2$
 $= -(x^2 - 3x - 4)$
 $\therefore y = -(x-4)(x+1)$



✓ correct graphs.

The expression $\frac{|x| - 2}{4 + 3x - x^2} > 0$

when either $y = |x| - 2$ and $y = 4 + 3x - x^2$ are both positive, or they are both negative.

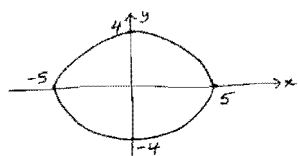
From the graphs we see that this is

when: $-2 < x < -1$ or $2 < x < 4$.

✓✓

Q5

(a) (i) $16x^2 + 25y^2 = 400$
 $\therefore \frac{x^2}{25} + \frac{y^2}{16} = 1$



(ii) $b^2 = a^2(1 - e^2)$
 $\therefore 1 - e^2 = \frac{b^2}{a^2}$
 $= \frac{16}{25}$
 $\therefore e^2 = \frac{9}{25}$
 $e = \frac{3}{5}$

(iii) $16x^2 + 25y^2 = 400$

$\therefore 32x + 50y \cdot \frac{dy}{dx} = 0$
 $\frac{dy}{dx} = -\frac{32x}{50y}$
 $= -\frac{16x_1}{25y_1}$ at $P(x_1, y_1)$

$\therefore m_N \text{ at } P = \frac{25y_1}{16x_1}$

Thus normal is:

$y - y_1 = \frac{25y_1}{16x_1} (x - x_1)$

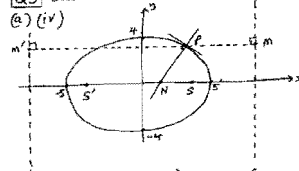
$16x_1y - 16x_1y_1 = 25y_1x - 25x_1y_1$

$\therefore 25y_1x - 16x_1y - 9x_1y_1 = 0$
 as required.

✓ for m_N at P

✓ correct substitution and expansion

Q5 - ctd.



$N = \left(\frac{9x_1}{25}, 0\right)$ $S = (3, 0)$
 $S' = (-3, 0)$
 Directrices are $x = \pm \frac{25}{3}$

Thus $\frac{NS}{NS'} = \frac{3 - \frac{9x_1}{25}}{\frac{9x_1}{25} + 3}$
 $= \frac{25 - 3x_1}{25 + 3x_1}$

Now $\frac{PS}{PM} = \frac{PS'}{PM'} = e$ (by definition)

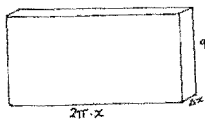
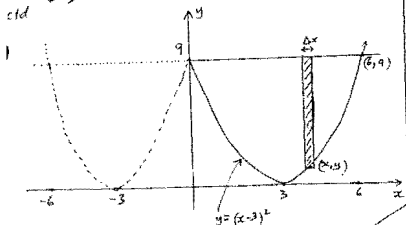
$\therefore \frac{PS}{PS'} = \frac{PM}{PM'}$
 $= \frac{25 - x_1}{x_1 + \frac{25}{3}}$
 $= \frac{25 - 3x_1}{25 + 3x_1}$
 $= \frac{NS}{NS'}$, as required.

3 marks Correct solution

2 marks Uses focus-directrix definition or other correct method AND obtains coords of N, S, S' , $x = \pm \frac{25}{3}$

1 mark Uses focus-directrix definition or other correct method OR obtains coords of N, S, S' , $x = \pm \frac{25}{3}$

5(b)



$$\begin{aligned}
 A(x) &= 2\pi x(9-y) \\
 \therefore \Delta V &\approx 2\pi x(9-y)\Delta x \\
 \text{But } 9-y &= 9-(x-3)^2 \\
 &= 6x-x^2 \\
 \therefore V &= \lim_{\Delta x \rightarrow 0} \sum_{x=0}^6 2\pi x(6x-x^2)\Delta x \\
 &= 2\pi \int_0^6 (6x^2-x^3) dx \\
 &= 2\pi \left[2x^3 - \frac{x^4}{4} \right]_0^6 \\
 &= 2\pi (432 - 324) \\
 \therefore V &= 216\pi \text{ units}^3
 \end{aligned}$$

(b) 4 marks: Correct solution -

3 marks:

$$\begin{aligned}
 \Delta V &= 2\pi x(9-y)\Delta x \\
 9-y &= 6x-x^2 \\
 V &= 2\pi \int_0^6 (6x^2-x^3) dx
 \end{aligned}$$

OR Correctly uses $y=(x-3)^2$ to change $y \rightarrow x$ but makes ONE error in the subsequent working.

2 marks:

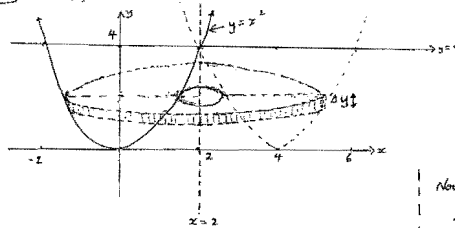
2 errors

1 mark:

Correctly finds definite integral from incorrect expression for V

OR 1 step of correct working

Q5-ctd. (c)



Consider a slice in the form of a washer perpendicular to the line $x=2$.

It has outer radius $x+2$, and inner radius $2-x$, and thickness Δy .

The volume is thus:

$$\Delta V = \pi[(2+x)^2 - (2-x)^2] \Delta y$$

$$\begin{aligned}
 \text{Now } y &= x^2 \therefore x = \sqrt{y} \\
 \therefore \Delta V &= \pi[(2+\sqrt{y})^2 - (2-\sqrt{y})^2] \Delta y \\
 &= \pi(8\sqrt{y}) \Delta y \\
 \therefore V &= \pi \int_0^4 8\sqrt{y} dy \\
 &= 8\pi \left[\frac{2}{3} y^{3/2} \right]_0^4 \\
 &= 8\pi \cdot \frac{2}{3} \cdot 8 \\
 \therefore V &= \frac{128\pi}{3} \text{ units}^3
 \end{aligned}$$

Marks: (c) 4 marks

Correct solution leading to answer of $\frac{128\pi}{3}$ units³

3 marks

Correct $\Delta V = \pi[(2+x)^2 - (2-x)^2] \Delta y$

Correct change of $x \rightarrow y$

Correct limits

Correct $V = \pi \int_0^4 8\sqrt{y} dy$

OR Incorrect ΔV involving annulus with $R_1, R_2 \neq 2+x$ and $2-x$

Correct change of $x \rightarrow y$

Correct limits

Correct answer from previous error

2 marks

Incorrect ΔV not involving an annulus following through with no further error

OR Incorrect ΔV involving an annulus with $R_1, R_2 \neq 2+x$ or $2-x$ following through with no further error

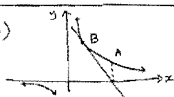
1 mark

Correct change of $x \rightarrow y$

OR Correct definite integral from & clear incorrect expression for V

OR Annulus with R_1, R_2

Q6 (a)



We have $B = (ct_2, \frac{c}{t_2})$ & $A = (ct_1, \frac{c}{t_1})$

$$A: y = c^2 x^{-1}$$

$$\therefore \frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\therefore m_T = -\frac{c^2}{ct_2^2}$$

$$= -\frac{1}{t_2^2}$$

$$\therefore \text{Tangent at B is: } y - \frac{c}{t_2} = -\frac{1}{t_2^2}(x - ct_2)$$

$$\therefore t_2^2 y - ct_2 = -x + ct_2$$

At the foot of A, $y=0 \therefore x=2ct_2$

But this equals the x-coordinate of A.

$$\text{i.e. } ct_1 = 2ct_2$$

$$\therefore t_1 = 2t_2 \text{ as required.}$$

$$\text{i) midpoint of AB} = \left(\frac{c}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right), \frac{c}{2} \left(\frac{1}{t_1} + \frac{1}{t_2} \right) \right)$$

$$\therefore \text{From part (i), midpoint} = \left(\frac{c}{2} (3t_2), \frac{c}{2} \left(\frac{1}{2t_2} + \frac{1}{t_2} \right) \right)$$

$$= \left(\frac{3ct_2}{2}, \frac{3c}{4t_2} \right)$$

Eliminating the parameter t_2 , we get:

$$xy = \frac{3ct_2}{2} \cdot \frac{3c}{4t_2}$$

$$\therefore xy = \frac{9c^2}{8}$$

which represents a rectangular hyperbola.

(i) 3 marks: Correct solution

2 marks: Finds equation of tangent at B

OR Finds line for A, C, $\frac{1}{t_1}$, $\frac{1}{t_2}$

1 mark: Finds gradient of tangent at B

OR Recognises x-intercept is x coord of A

OR A, C, $\frac{1}{t_1}$, $\frac{1}{t_2}$

(ii) 2 marks: correct solution

1 mark: correct midpoint in terms of t_2

Q6-ctd.

$$(b) (i) |z-2| = 2 \left(\operatorname{Re} z - \frac{1}{2} \right)$$

$$\text{Let } z = x + iy$$

$$\text{Since } |z-2| \geq 0, \therefore \operatorname{Re} z \geq \frac{1}{2}, \text{ i.e. } x \geq \frac{1}{2}$$

We have:

$$|(x-2) + iy| = 2(x - \frac{1}{2})$$

$$\sqrt{(x-2)^2 + y^2} = 2(x - \frac{1}{2})$$

$$(x-2)^2 + y^2 = 4(x - \frac{1}{2})^2$$

$$\therefore x^2 - 4x + 4 + y^2 = 4x^2 - 4x + 1$$

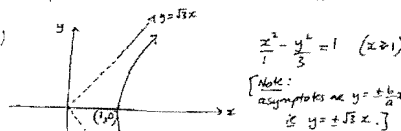
$$\therefore 3x^2 - y^2 = 3$$

$$\therefore \frac{x^2}{1} - \frac{y^2}{3} = 1 \quad (x \geq 1)$$

[Alternatively, candidates could use the focus-directrix definition with:

$$c=2, S=(2,0), \text{directrix } x=\frac{1}{2}.$$

(ii)



Thus:

$$|z| \geq 1$$

$$-\frac{\pi}{3} < \arg z < \frac{\pi}{3}$$

(b)(i) 3 marks: Correct solution

2 marks: Correct derivative of equation

OR correct branch and correct expansion of LHS

OR correct branch and correct application of AB

1 mark: correct branch

OR correct expansion $\Rightarrow 3x^2 - y^2 = 3$

OR correct expansion of RHS or LHS

(ii) 3 marks: correct solution

1 mark for each of:
 - correct sketch
 - must show asymptotes
 - or indicate a
 - or other clear indication
 - graph is a hyperbola

Q6 (c)

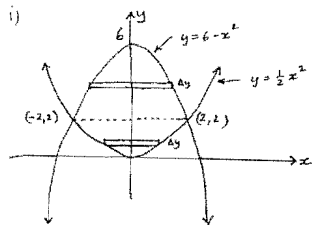
At intersections, $\frac{1}{2}x^2 = 6 - x^2$.

$$x^2 = 12 - 2x^2$$

$$3x^2 = 12$$

$$x = \pm 2$$

Thus intersections are $(-2, 2)$ and $(2, 2)$



Area of semi-circular disks = $\frac{1}{2} \pi x^2$

But we must express this in terms of y , but the expression will depend on whether disks are above or below the line $y=2$.

Above: $x^2 = 6 - y$, Below: $x^2 = 2y$

$$\text{Thus: } V = \frac{\pi}{2} \int_0^2 2y dy + \frac{\pi}{2} \int_2^6 (6-y) dy$$

$$= \frac{\pi}{2} [y^2]_0^2 + \frac{\pi}{2} [6y - \frac{y^2}{2}]_2^6$$

$$= \frac{\pi}{2} [4-0] + \frac{\pi}{2} [(36-18) - (12-2)]$$

$$= \frac{\pi}{2} (4 + 18 - 10)$$

$$\therefore V = 6\pi \text{ units}^3.$$

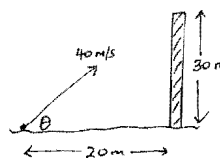
⑥ (a) (i) mark: correct solution

(ii) 3 marks: correct solution
2 marks: correct integrals with limits
or wrong limits with no further error

1 mark: correct dv
or 1 correct integral with limits
or use $\int x^n = \frac{x^{n+1}}{n+1}$ with no further errors

Q7

(a) (i)



$$\ddot{x} = 0$$

$$\therefore \dot{x} = c$$

$$\text{at } t=0, \dot{x} = 40 \cos \theta \therefore c = 40 \cos \theta$$

$$\therefore x = 40 \cos \theta$$

$$\therefore x = 40 \cos \theta t + c$$

$$\text{at } t=0, x=0 \therefore c=0$$

$$\therefore x = 40 \cos \theta \cdot t$$

$$\ddot{y} = -10$$

$$\therefore \dot{y} = -10t + c$$

$$\text{at } t=0, \dot{y} = 40 \sin \theta \therefore c = 40 \sin \theta$$

$$\therefore y = -10t^2 + 40 \sin \theta$$

$$\therefore y = -5t^2 + 40 \sin \theta t + c$$

$$\text{at } t=0, y=0 \therefore c=0$$

$$\therefore y = -5t^2 + 40 \sin \theta \cdot t$$

✓ derives x equation

✓ derives y equation

(ii) Cartesian equation of path is:

$$y = -5 \left(\frac{x}{40 \cos \theta} \right)^2 + 40 \left(\frac{x}{40 \cos \theta} \right) \sin \theta$$

at $x=20, y>30$:

$$\therefore -5 \left(\frac{20}{40 \cos \theta} \right)^2 + 40 \left(\frac{20}{40 \cos \theta} \right) \sin \theta > 30$$

$$\therefore -\frac{5}{4} \sec^2 \theta + 20 \tan \theta - 30 > 0$$

$$\therefore -(1 + \tan^2 \theta) + 16 \tan \theta - 24 > 0$$

$$\therefore \tan^2 \theta - 16 \tan \theta + 25 < 0$$

$$(\tan \theta - 16 \tan \theta + 64) - 39 < 0$$

$$(\tan \theta - 8)^2 < 39$$

$$\therefore -\sqrt{39} < \tan \theta - 8 < \sqrt{39}$$

$$\therefore -\sqrt{39} + 8 < \tan \theta < \sqrt{39} + 8$$

$$\therefore 60^\circ 19' < \theta < 85^\circ 59'$$

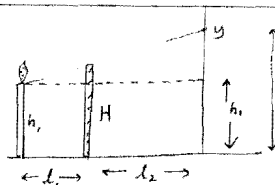
✓ correct trajectory & use of data

✓ correct simplification

✓ correct tan theta

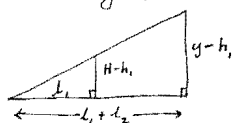
✓ correct angles range

Q7 ctd (b)



The goal is to find $\frac{dy}{dt}$. (Note: H, l_1, l_2 are all constants)

Using similar triangles, we have:-



$$\text{Thus } \frac{y - h_1}{l_1 + l_2} = \frac{H - h_1}{l_1}$$

$$\therefore y - h_1 = \frac{l_1 + l_2}{l_1} (H - h_1)$$

$$\therefore y = \frac{l_1 + l_2}{l_1} (H - h_1) + h_1$$

$$\text{Thus } \frac{dy}{dt} = \frac{l_1 + l_2}{l_1} \cdot -1 \cdot \frac{dh_1}{dt} + \frac{dh_1}{dt}$$

$$\text{But } \frac{dh_1}{dt} = -3$$

$$\therefore \frac{dy}{dt} = 3 \left(\frac{l_1 + l_2}{l_1} \right) - 3$$

$$(\text{i.e. } \frac{dy}{dt} = 3 \left[\frac{l_1 + l_2}{l_1} - 1 \right])$$

$$= 3 \cdot \frac{l_2}{l_1}$$

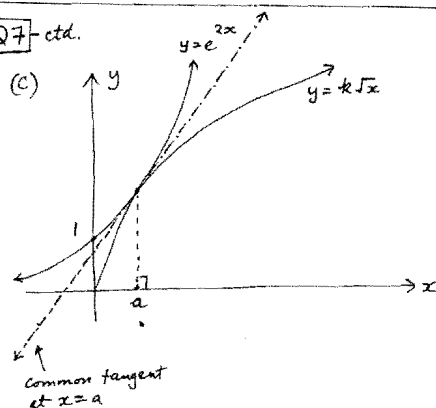
✓ equation relating y to h_1, l_1, l_2, H .

✓ correct derivative

✓ final correct relation (any correct form is OK)

Q7 ctd.

(c)



If the curves $y = e^{2x}$ and $y = k\sqrt{x}$ intersect just once, at $x=a$ (say) then we know two facts:

(1) they have the same y -value at $x=a$ and, (2) they share a tangent line at $x=a$

$$\text{Thus: (1) } e^{2a} = k\sqrt{a}$$

$$\text{and (2) } 2e^{2a} = \frac{k}{2\sqrt{a}}$$

Solving simultaneously,

$$2 \cdot k\sqrt{a} = \frac{k}{2\sqrt{a}}$$

$$\therefore a = \frac{1}{4}$$

$$\text{Thus } e^{2(\frac{1}{4})} = k\sqrt{\frac{1}{4}}$$

$$\therefore e^{\frac{1}{2}} = k \cdot \frac{1}{2}$$

$$\therefore k = 2\sqrt{e}$$

✓ correct reasoning / realisation they share a tangent line

✓ correct working

✓ correct answer

17] ctd.

$$f(x) = \sqrt{8x - x^2} - \sqrt{14x - x^2 - 48}$$

$$= \sqrt{16 - (x-4)^2} - \sqrt{1 - (x-7)^2}$$

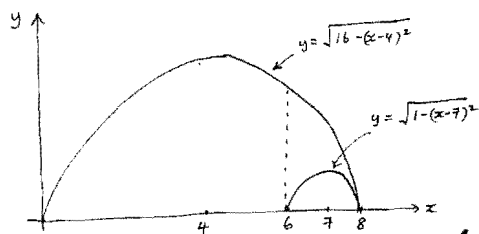
Thus $f(x)$ is the difference of two functions: $y = \sqrt{16 - (x-4)^2}$ & $y = \sqrt{1 - (x-7)^2}$.

We can graph these on the same number plane:

$y = \sqrt{16 - (x-4)^2}$ is the top half of the circle $(x-4)^2 + y^2 = 16$, and has D: $0 \leq x \leq 8$

$y = \sqrt{1 - (x-7)^2}$ is the top half of the circle $(x-7)^2 + y^2 = 1$, and has D: $6 \leq x \leq 8$.

Graphing them, we see:



In the domain of $f(x)$, D: $6 \leq x \leq 8$, the maximum difference occurs at $x=6$.

Thus maximum value of $f(x)$ is equal to $f(6) = \sqrt{12}$.

Correct answer with correct working, & using graph(s) : 3

Substantially correct (max. of 1 error), & including graph(s) : 2

Some correct reasoning (and including a correct graph) : 1

Q8 (a)

$$I_n = \int_0^{\pi/2} \sin^n x \, dx$$

$$= \int_0^{\pi/2} \sin^{n-1} x \cdot \frac{d}{dx}(-\cos x) \, dx$$

$$= \left[-\cos x \cdot \sin^{n-1} x \right]_0^{\pi/2} - \int_0^{\pi/2} (-\cos x) \cdot (n-1) \sin^{n-2} x \cdot \cos x \, dx$$

$$= 0 + (n-1) \int_0^{\pi/2} \cos^2 x \cdot \sin^{n-2} x \, dx$$

$$= (n-1) \int_0^{\pi/2} (1 - \sin^2 x) \cdot \sin^{n-2} x \, dx$$

$$= (n-1) \int_0^{\pi/2} (\sin^{n-2} x - \sin^n x) \, dx$$

$$\therefore I_n = (n-1) [I_{n-2} - I_n]$$

$$\therefore I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{n-1}{n} \cdot I_{n-2} \text{ as required.}$$

$$\text{Now, } \therefore I_4 = \frac{3}{4} I_2$$

$$= \frac{3}{4} \cdot \frac{1}{2} \cdot I_0$$

$$= \frac{3}{4} \cdot \frac{1}{2} \cdot \int_0^{\pi/2} dx$$

$$= \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$\therefore I_4 = \frac{3\pi}{16}$$

$$\text{So } \therefore I_6 = \frac{5}{6} \cdot I_4$$

$$\therefore I_6 = \frac{15\pi}{96}$$

$$\therefore I_6 = \frac{5\pi}{32}$$

✓ correct Integration by parts.

✓ correct simplification

✓ correctly simplified

✓ correct I_4

✓ correct I_6

28] ctd. (b)

(i) If $x = a \cos^3 \theta$ & $y = a \sin^3 \theta$, then

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} \cos^2 \theta + a^{\frac{2}{3}} \sin^2 \theta$$

$$= a^{\frac{2}{3}} (\cos^2 \theta + \sin^2 \theta)$$

$$= a^{\frac{2}{3}}$$

So $\therefore$ it is a correct parametrisation.

✓ confirms parametrisation

$$(i) \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

$$= \frac{3a \sin^2 \theta \cdot \cos \theta}{3a \cos^2 \theta \cdot -\sin \theta}$$

$$= -\frac{\sin \theta}{\cos \theta}$$

$$\therefore \frac{dy}{dx} = -\tan \theta$$

So at P, $\frac{dy}{dx} = -\tan \phi$.

✓ derives $\frac{dy}{dx}$ correctly

(i) The tangent at P is given by:

$$y - a \sin^3 \phi = -\tan \phi (x - a \cos^3 \phi)$$

$$= -\frac{\sin \phi}{\cos \phi} (x - a \cos^3 \phi)$$

$$\therefore \cos \phi y - a \sin^3 \phi \cos \phi = -\sin \phi x + a \sin \phi \cos^3 \phi$$

$$\therefore \sin \phi \cdot x + \cos \phi \cdot y = a \sin \phi \cos \phi$$

✓ A correct equation of the tangent

Thus for y-intercept ($x=0$), $y = a \sin \phi$
& for x-intercept ($y=0$), $x = a \cos \phi$.

✓ correct intercepts

Thus length cut off by axes is

$$\text{given by } \sqrt{(a \sin \phi)^2 + (a \cos \phi)^2}$$

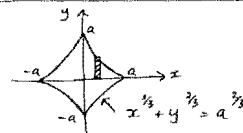
$$= \sqrt{a^2 (\sin^2 \phi + \cos^2 \phi)}$$

$$= a, \text{ which is a constant.}$$

✓ shows the intercepted length is constant.

Q8] ctd

(b) (iv)



Let A = the area enclosed in Quadrant 1.

$$\text{Then } A = \int_0^a y \, dx$$

$$= \int_0^a (a^{\frac{2}{3}} - x^{\frac{2}{3}})^{\frac{3}{2}} \, dx$$

Using a substitution $x = a \cos^3 \theta$, we see:

$$dx = 3a \cos^2 \theta \cdot -\sin \theta \cdot d\theta$$

$$\text{Also, when } \begin{cases} x=0, \cos^3 \theta = 0 \therefore \theta = \frac{\pi}{2} \\ x=a, \cos^3 \theta = 1 \therefore \theta = 0 \end{cases}$$

$$\text{Thus, } A = \int_{\frac{\pi}{2}}^0 (a^{\frac{2}{3}} - a^{\frac{2}{3}} \cos^2 \theta)^{\frac{3}{2}} \cdot -3a \cos^2 \theta \sin \theta \, d\theta$$

$$= 3a \int_0^{\frac{\pi}{2}} a (1 - \cos^2 \theta)^{\frac{3}{2}} \cdot \cos^2 \theta \sin \theta \, d\theta$$

$$= 3a^2 \int_0^{\frac{\pi}{2}} \sin^3 \theta \cdot \cos^2 \theta \sin \theta \, d\theta$$

$$= 3a^2 \int_0^{\frac{\pi}{2}} \sin^4 \theta (1 - \sin^2 \theta) \, d\theta$$

$$= 3a^2 \int_0^{\frac{\pi}{2}} (\sin^4 \theta - \sin^6 \theta) \, d\theta$$

But from part (a) of Q8,

$$\int_0^{\frac{\pi}{2}} (\sin^4 \theta - \sin^6 \theta) \, d\theta = I_4 - I_6$$

$$= \frac{3\pi}{16} - \frac{5\pi}{32}$$

$$= \frac{\pi}{32}$$

$$\therefore A = 3a^2 \cdot \frac{\pi}{32} = \frac{3a^2 \pi}{32}$$

$$\therefore \text{area of astroid} = 4A = \frac{3a^2 \pi}{8} \text{ units}^2$$

✓ correct Cartesian integral

✓ correct substitution

✓ correct simplifications

✓ correct evaluation of integrals

✓ correct final answer.